

**NORME
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**CEI
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60897

Première édition
First edition
1987-03

**Méthodes de détermination de la tension de
claquage au choc de foudre des liquides isolants**

**Methods for the determination of the lightning
impulse breakdown voltage of insulating liquids**



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International Electrotechnical Commission
Telefax: +41 22 919 0300

3, rue de Varembe Geneva, Switzerland
e-mail: inmail@iec.ch IEC web site <http://www.iec.ch>



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INTERNATIONAL ELECTROTECHNICAL COMMISSION

**METHODS FOR THE DETERMINATION
OF THE LIGHTNING IMPULSE BREAKDOWN VOLTAGE
OF INSULATING LIQUIDS**

FOREWORD

- 1) The formal decisions or agreements of the IEC on technical matters, prepared by Technical Committees on which all the National Committees having a special interest therein are represented, express, as nearly as possible, an international consensus of opinion on the subjects dealt with.
- 2) They have the form of recommendations for international use and they are accepted by the National Committees in that sense.
- 3) In order to promote international unification, the IEC expresses the wish that all National Committees should adopt the text of the IEC recommendation for their national rules in so far as national conditions will permit. Any divergence between the IEC recommendation and the corresponding national rules should, as far as possible, be clearly indicated in the latter.

PREFACE

This standard has been prepared by IEC Sub-Committee 10A: Hydrocarbon Insulating Oils, of IEC Technical Committee No. 10: Fluids for Electrotechnical Applications.

The text of this standard is based on the following documents:

Six Months' Rule	Report on Voting
10A(CO)64	10A(CO)68

Further information can be found in the Report on Voting indicated in the table above.

METHODS FOR THE DETERMINATION OF THE LIGHTNING IMPULSE BREAKDOWN VOLTAGE OF INSULATING LIQUIDS

1. Scope

- 1.1 This standard describes two test methods, A and B, for assessing the electric strength of insulating liquids in a divergent field when subjected to standard lightning impulse.

Method A is based on a step procedure intended to provide an estimate of impulse breakdown voltage under specific conditions.

Method B is a statistical test designed to check an hypothesis about the impulse breakdown probability of an insulating liquid at a given voltage level.

- 1.2 Both methods can be applied to unused or used insulating liquids, the viscosity of which is lower than 700 mm²/s at 40 °C.

The methods can be used with positive or negative impulses. No specific stipulations are made regarding the preparation of the liquid sample provided that it corresponds to industrial practice; however, tests made before and after treatment of the sample may be useful for demonstrating the effect of treatment.

- 1.3 Both methods are primarily intended to establish standardized procedures for assessing the impulse electric strength of insulating liquids. They may serve to differentiate dielectric liquids between each other and to detect variations in their characteristics due to modifications in their chemical composition as the result of changes in the manufacturing process and/or feed-stocks.

2. General

- 2.1 Insulating liquids in service in electrical apparatus may be subjected to switching or lightning transient voltage stress superimposed on continuous operation voltage at power frequency.

Whether such surges be unidirectional or oscillatory, the result will always be a transient operation with positive or negative polarity, requiring knowledge of the behaviour of the insulating liquid under these conditions.

However, much more experience is necessary before the results of impulse breakdown voltage obtained in test cells employing point-to-sphere electrode geometry can be correlated with performance of the liquid in practical insulation systems.

- 2.2 In insulating liquids, impulse breakdown is a complex phenomenon, still not fully understood; it requires the initiation and the propagation of a prebreakdown disturbance (streamers).

Breakdown voltage appears to be determined by such factors as voltage wave shape, duration of the applied voltage and field configuration.

To obtain comparable results it is necessary for all these factors to be clearly defined and kept under close control. Even so, however, a spread of results is frequently observed which is believed to be associated with the random nature of prebreakdown mechanisms.

- 2.3 Whilst breakdown behaviour in a symmetrical field is not affected by the polarity of voltage applied, polarity does have a marked effect in divergent fields, particularly in point-to-sphere geometry. In this configuration, experience has shown that the chemical composition of the liquid has a

fundamental role in determining the negative impulse breakdown properties.

Highly divergent field geometries, as used in the methods described, are therefore necessary to differentiate composition effects in insulating liquids.

- 2.4 Impulse breakdown voltage depends on wave front duration. Consequently, the methods described specify only the standard full impulse wave (1.2/50).
- 2.5 Unlike power frequency breakdown voltage (IEC Publication 156), the point-to-sphere impulse breakdown is largely independent of such contaminants as moisture and particles. Accordingly, no specific provisions are made to control these contaminants, provided their concentrations do not exceed the serviceability limits of the liquid.

3. Documents referred to

IEC Publications:

- 52 (1960): Recommendations for Voltage Measurement by means of Sphere-gaps (One Sphere Earthed).
- 60-1 (1973): High-voltage Test Techniques, Part 1: General Definitions and Test Requirements.
- 60-2 (1973): Part 2: Test Procedures.
- 60-3 (1976): Part 3: Measuring Devices.
- 60-4 (1977): Part 4: Application Guide for Measuring Devices.
- 71-1 (1976): Insulation Co-ordination, Part 1: Terms, Definitions, Principles and Rules.
- 71-2 (1976): Part 2: Application Guide.
- 156 (1963): Method for the Determination of the Electric Strength of Insulating Oils.
- 475 (1974): Method of Sampling Liquid Dielectrics.

ISO Standards:

- 2854 (1976): Statistical Interpretation of Data — Techniques of Estimation and Tests relating to Means and Variances.
- 3534 (1977): Statistics—Vocabulary and Symbols.
- 5725 (1986): Precision of Test Methods—Determination of Repeatability and Reproducibility by Interlaboratory Tests.

4. Apparatus

4.1 *Impulse generator*

The impulse generator shall be capable of producing a standard 1.2/50 full wave, adjustable to positive or negative polarity in accordance with IEC Publication 60-1 and with accuracy as defined in IEC Publication 60-2; in particular, measurement to within $\pm 3\%$ of the crest voltage is necessary. The generator shall have a voltage rating of at least 300 kV and output energies in the range of 0.1 kJ to 20 kJ are appropriate.

4.2 *Adjustment of impulse voltage*

This operation is very important. The crest voltage should therefore be predetermined to within 1% by a manual control charging apparatus or, preferably, by means of an automatically triggered apparatus with a regulating accuracy of $\pm 0.5\%$.

4.3 *Impulse voltage measurement*

The impulse voltage measurement should be carried out as laid down by IEC Publications 60-3 and 60-4. An accurately calibrated resistive voltage divider and a crest voltmeter are recommended in preference to an oscilloscope. However, calibration of the measuring system by sphere-gap in accordance with IEC Publication 52 is acceptable. The error in measurement of the crest value of the impulse voltage should be known and should not exceed 3%.

4.4 *Test-cell design*

- 4.4.1 The cell consists of a vessel containing a vertical gap as shown for example in Figure 1, page 26. The volume of liquid in this cell is of the order of 300 ml.

Metal parts shall be restricted to electrodes and their supports.

- 4.4.2 The test-cell should be designed for ease of dismantling and thorough cleaning. Dimensions should ensure a flashover voltage of at least 250 kV.

- 4.4.3 Insulating materials used in the cell shall be of high dielectric strength, thermally stable for temperatures up to 80 °C, compatible with insulating liquids to be tested, and resistant to solvents and cleaning agents commonly used for these liquids.

- 4.4.4 The gap shall be composed of two electrodes forming an adjustable point-to-sphere configuration. The spherical electrode shall be of polished steel and may consist of a bearing ball 12.5 mm to 13 mm in diameter. This ball is held in place by a magnet. The point electrode shall be a gramophone needle, with a conical tip having a radius of curvature in the range 40 µm to 70 µm. The shape of the needle and the radius of curvature should be checked using a microscope; Appendix B gives a method for measuring the radius of curvature.

5. *Sampling of the liquid*

Samples of the insulating liquid to be tested shall be taken in accordance with IEC Publication 475.

6. *Preparation and maintenance of the cell*

6.1 *Cell cleaning*

All parts of the cell, including the spherical electrode and gramophone needle, shall be de-greased with reagent-grade heptane, washed with detergent, rinsed thoroughly with hot tap-water and then with distilled water.

The parts should then be dried with oil-free dehydrated compressed air, and kept in a desiccator until required.

6.2 *Daily use*

The cell prepared as described in Sub-clause 6.1 is ready for the tests. Before testing a new sample, repeat the prescribed cleaning using an appropriate solvent.

As long as the tests are applied to the same sample, rinsing out with the liquid between each filling is adequate.

7. *Preparation of test*

- 7.1 Rinse the cell and electrodes thoroughly with the sample liquid. Fill the test-cell, slowly, making sure that there are no gas bubbles. Allow the liquid to settle for at least 5 min before testing.

7.2 Electrode spacing

Gently set the electrodes in contact, checking contact with an ohmmeter; then displace one of the electrodes to the desired spacing value using a dial micrometer, a helical device or a thickness gauge. The gap spacing shall be adjusted to the values defined in Sub-clause 8.1.2, with a tolerance of 0.1 mm.

- 7.3 Earth the spherical electrode; the connection shall be as short as possible. The impulse generator output shall be applied to the point electrode and care must be taken to avoid too large a connection loop.
- 7.4 The temperature of the sample when tested should be the same as the temperature of the test-room; a room temperature between 15 °C and 30 °C is appropriate. Record the test temperature.

8. Test procedures

8.1 Method A—Step test

8.1.1 Principle

1.2/50 standard lightning impulse voltages of increasing crest value are applied to the liquid specimen using a point-to-sphere electrode system until breakdown occurs. Five measurements are carried out and their average is taken as the lightning impulse breakdown voltage of the liquid tested.

The initial value of the test voltage, the voltage step and the electrode spacing depend on the breakdown voltage value of the liquid tested.

8.1.2 Procedure

- a) Prepare the test-cell as described in Clause 7.
- b) From Table I, select the appropriate initial value of the test voltage (U_i) voltage step and gap spacing, based on the expected breakdown (U_0) at 15 mm gap spacing.
- c) Apply one impulse (of the selected polarity) at the initial selected voltage and then increase the voltage in steps until breakdown occurs. One impulse shall be applied at each voltage level, allowing a minimum of 1 min. between two successive impulses.
- d) The procedure as described in Items a), b), c) shall be repeated until five breakdown values have been obtained for the liquid tested. After each breakdown the point electrode is changed and the sphere electrode rotated, then follow Sub-clause 7.1 and 7.2. The sphere electrode shall be changed every five breakdowns.
- e) For the test to be valid, the test sample shall withstand a minimum of three levels before breakdown occurs. If breakdown occurs before, repeat the test with a lower initial voltage, 5 kV or 10 kV according to the case.
- f) Note the prospective crest value of the impulse during which the breakdown occur as the nominal breakdown voltage.
- g) When the breakdown voltage of a liquid under test cannot be anticipated determine U_c following Items a) and c), using a gap of 15 mm, a starting voltage of 50 kV and a voltage step of 10 kV. Then continue as in Item a) to f).

If breakdown at 15 mm cannot be obtained below the flashover voltage of the cell (about 250 kV) decrease the gap spacing to 10 mm or, if necessary, 5 mm.

Amorphy

TABLE I
Selection of initial voltage and gap

Expected voltage at 15 mm (U_e (kV))		$50 \leq U_e \leq 100$	$100 < U_e \leq 250$	$U_e > 250$
Gap (mm)		25 ± 0.1	15 ± 0.1	10 ± 0.1
Initial voltage (U_i (kV))		$1.5 U_e - 25$	$U_e - 50$	150
Step voltage (kV)		5	5	10

8.1.3 Precision

The precision of a test method is characterized by its repeatability r and its reproducibility R (see ISO Standard 5725). These latter values for mineral insulating oils only, tested according to method A are given in Table II.

TABLE II
Repeatability and reproducibility of test method A for mineral transformer oil

Polarity of the impulses	r (%)	R (%)
negative	7	10
positive	15	30

Notes 1. — The values of r and R are expressed in per cent of the mean breakdown voltage.

2. — The values given in Table II are determined from results of tests performed on three different mineral insulating oils by seven laboratories.

When the method is correctly applied there is 5% probability that the difference between two "single results" (see note) obtained from two tests performed on the same oil under the same conditions (same operator, same apparatus, same laboratory and short interval of time), will exceed 7% of U in the case of negative impulses and 15% of U in the case of positive impulses, U being the average of the two results.

When two "single results" are obtained from tests performed on the same oil but under different conditions (different operators, different apparatus, different laboratories), there is 5% probability that the difference between two results will exceed 10% of U in the case of negative impulses and 30% of U in the case of positive impulses.

If the difference between the two results exceeds the value given in Table II, further action is required, for example verification of the apparatus and repetition of the tests.

Note. — A "single result" is the average of five breakdown voltages as specified in Sub-clause 8.1.1.

8.2 Method B—Sequential test

8.2.1 Principle

Experience shows that when an impulse wave is applied to the test-cell at a crest level close to the breakdown voltage measured by method A a breakdown may or may not occur. Thus the notion of breakdown probability P , which is a function of U but of unknown value, should be introduced. The sequential test permits the comparison of this breakdown probability with an arbitrary value P_0 and the testing hypothesis:

$H_0: P \leq P_0$, called the null hypothesis,
against the alternative hypothesis $H_1: P > P_0$ (see Appendix A).

The method consists in applying a sequence of impulses of constant crest value until breakdown occurs, and plotting the results on a ruling graph (see Appendix A and Figure 2, page 27).

The sequential test is carried on until a decision can be made; if no decision is possible the test will be terminated when a predetermined number of impulses is exceeded.

8.2.2 Procedure

- a) Draw the ruling graph for the appropriate values of P_0 and the parameters defined in Appendix A.
- b) Select a crest value U_0 for the impulse voltage and set the impulse generator (see Note 1).
- c) Prepare the test-cell as described in Clause 7, and set the electrode spacing to the desired value.
- d) Apply the first impulse of selected polarity and crest value. If there is no breakdown, wait 1 min. before applying another impulse, and then continue with further impulses until breakdown occurs.
- e) Whenever a breakdown occurs, change the point electrode, rotate the sphere electrode, refill the cell, and continue the procedure. Change the sphere electrode every five breakdowns.
- f) Plot each impulse and corresponding breakdown on the ruling graph (Figure 2).
- g) As long as the plot is within the area limited by the D_1 and D_2 lines (see Appendix A), no decision can be taken and the test must be continued (see Note 2).

If the plot intersects line D_1 , the null hypothesis H_0 is accepted: $P \leq P_0$.

If the plot intersects line D_2 , the null hypothesis H_0 is rejected: $P > P_0$.

Notes 1. — The test voltage U_0 may be chosen two voltage steps below the mean breakdown value found by method A.

2. — If after 85 impulses no decision can be made the test should be repeated at a level 5 kV or 10 kV lower.

9. Report

The report shall include the following items:

- 9.1 The number of this publication and the method applied.
- 9.2 Sample identification and preparation.
- 9.3 Gap spacing.

9.4 Method A

Starting voltage crest level, polarity and voltage step.
Impulse voltage for each breakdown.
Mean breakdown voltage.
Standard mean deviation.

9.5 Method B

Voltage crest level and polarity.
Selected statistical parameters.
Ruling graph and plot of results.

APPENDIX A

SEQUENTIAL TEST FOR COMPARISON OF THE BREAKDOWN PROBABILITY
OF LIQUID DIELECTRICS WITH A STANDARD VALUE

CONSTRUCTION OF A RULING GRAPH

A1. Introduction

This appendix outlines the principle of sequential test and provides instructions on the way the ruling graph is constructed. For a complete statement of sequential analysis see "Sequential Analysis" by Abraham WALD (J. Wiley and Sons Inc. Editor).

A2. List of symbols

The symbols employed are consistent with the definitions of International Standard ISO 3534.

$P(U)$ or P = Breakdown probability under standard lightning impulse of a dielectric liquid for a given value of the test voltage U (see note).

P_0 = Arbitrary fixed breakdown probability which may be related to accepted quality criteria.

P_1 = Breakdown probability lower than P_0 , the value of which is chosen to take into account the precision of the test voltage applied.

P_2 = Breakdown probability higher than P_0 and chosen for the same reasons as before.

H_0 = Null hypothesis: $P \leq P_0$.

H_1 = Alternative hypothesis. The hypothesis which is opposed to the null hypothesis, that is to say $P > P_0$.

α = Maximum risk of rejecting H_0 when $P \leq P_1$.

β = Maximum risk of accepting H_0 when $P \geq P_2$.

n = Number of impulses applied.

d_n = Number of breakdowns observed after n impulses have been applied.

A_n = Acceptance number.

R_n = Rejection number.

Note. — In every case P is unknown and the aim of the sequential test is not to determine this probability which would take too long tests, but to compare it with a fixed value P_0 with an accelerated procedure giving information as to satisfactory withstand levels in industrial practice.

A3. A statistical test is a rule permitting decision-making with accepted error risks from experimental results. It needs the definition of the hypothesis H_0 to be tested and the use of a function of these results given in this case by the points (d_n, n) in the ruling graph.

In the particular case of the sequential test, the function of the experimental results takes values in three mutually exclusive subsets of the set of these results, defining as the test proceeds either the acceptance of H_0 or the rejection of H_0 or the continuation of the test.

- A4. Let P_0 be an arbitrary value. We wish to test the hypothesis $H_0: P \leq P_0$ against the alternative hypothesis $H_1: P > P_0$ at a value U of the test voltage applied.

For practical considerations a value P_1 lower than P_0 and an error risk α are fixed such that the probability of rejecting H_0 when $P \leq P_1$ is lower or equal to α .

Likewise a value P_2 higher than P_0 and an error risk β are fixed such that the probability of accepting H_0 when $P \geq P_2$ is lower or equal to β .

- A5. Let L be the likelihood ratio defined as $L = L_2/L_1$ where L_1 is the probability of obtaining a set of observed values with $P = P_1$ and L_2 is the probability of obtaining a set of observed values with $P = P_2$.

The sequential test consists in calculating L at any stage of the experiment and in taking the following decision rules:

If $L \leq A$, H_0 is accepted

If $L \geq R$, H_0 is rejected

If $A < L < R$, the experiment is continued by making an additional observation.

A and R are chosen in such a way that the probability of rejecting H_0 when $P = P_1$ is α and the probability of accepting H_0 when $P = P_2$ is β .

An approximate calculation shows that:

$$A = \beta / (1 - \alpha)$$

$$R = (1 - \beta) / \alpha$$

To summarize, if $P \leq P_1$ the probability of rejecting H_0 is always lower or equal to α and if $P \geq P_2$ the probability of accepting H_0 is always lower or equal to β .

- A6. The probability L_1 that d_n breakdowns would occur in n impulse applications with $P = P_1$ is:

$$L_1 = \binom{n}{d_n} P_1^{d_n} (1 - P_1)^{n-d_n}$$

The probability L_2 that d breakdowns would occur in n impulse applications with $P = P_2$ is:

$$L_2 = \binom{n}{d_n} P_2^{d_n} (1 - P_2)^{n-d_n}$$

Therefore the likelihood ratio is:

$$L = P_2^{d_n} (1 - P_2)^{n-d_n} / P_1^{d_n} (1 - P_1)^{n-d_n}$$

Suppose that after n impulses the conditions $L \leq A$ is fulfilled then:

$$P_2^{d_n} (1 - P_2)^{n-d_n} / P_1^{d_n} (1 - P_1)^{n-d_n} \leq \beta / (1 - \alpha)$$

The above expression yields the following solution for the observed number of discharges d_n : $d_n \leq h_1 + sn$ where h_1 and s are constants defined by the following expressions:

$$h_1 = \log \frac{\beta}{1 - \alpha} / \left(\log \frac{P_2}{P_1} - \log \frac{1 - P_2}{1 - P_1} \right)$$

$$s = \log \frac{1 - P_2}{1 - P_1} / \left(\log \frac{P_2}{P_1} - \log \frac{1 - P_2}{1 - P_1} \right)$$

Let

$$A_n = h_1 + sn \quad (1)$$

When $d_n \leq A_n$ condition $L \leq A$ is fulfilled and the null hypothesis H_0 is accepted. Therefore A_n can be called the acceptance number.

- A7. Suppose now that after n impulses the condition $L \geq R$ is fulfilled. Proceeding in the same way as before the following expression is obtained for d_n : $d_n \geq h_2 + sn$, where s has the same meaning as above and h_2 is given by:

$$h_2 = \log \frac{1-\beta}{\alpha} / \left(\log \frac{P_2}{P_1} - \log \frac{1-P_2}{1-P_1} \right)$$

$$\text{Let: } R_n = h_2 + sn \quad (2)$$

When $d_n \geq R_n$ condition $L \geq R$ is fulfilled and the null hypothesis H_0 is rejected. Therefore R_n can be called the rejection number.

- A8. In the plane of coordinates n and d , the result of the test after n impulses, can be represented by the point (n, d_n) . A_n and R_n (see (1) and (2) above) are the equations of two parallel lines (D_1) and (D_2) respectively, which divide the plane in three areas (see ruling graph, Figure 2, page 27).

If the point (n, d_n) is in the area below the line (D_1) , H_0 is accepted and if the point is in the area above the line (D_2) , H_0 is rejected. If the point is between the two lines no decision is taken and another impulse is applied.

- A9. The ruling graph shown in Figure 2 is an example drawn from the following values:

$$P_0 = 10\%; P_1 = 5\%; P_2 = 15\%; \alpha = 5\% \text{ and } \beta = 1\%.$$

- A10. Unlike the conventional statistical test, the equivalent sequential test does not require a predetermined number of tests be performed. The number of tests is a random variable on average lower than the number of tests required by the equivalent conventional statistical test.

Moreover, the probability that the sequential test will be carried on indefinitely is demonstrated to be nil. However, in order to keep the test length within limits the procedure can be terminated using either a truncating rule or a decision according to Sub-clause 8.2.2.

APPENDIX B

MEASUREMENT OF RADIUS OF CURVATURE OF NEEDLES

- B1. This appendix describes a method for determining the radius of curvature of the point electrode by a metallographic microscope.

Figure 3, page 27, gives the schematic diagram of the optical system.

The light emitted by the source of light of the microscope (S) is reflected by a set of mirrors at an angle of 45° (M) on the object (A) to be examined. The image is then projected, through lenses (L1, L2), on a ground glass plate (D). Settings are made so that the object is finally magnified a thousand times.

Using a scale graduated in millimetres, measure directly on the ground glass plate (D) the sagitta (a) corresponding to chord (b).

For greater accuracy, make two measurements corresponding to two chords of different values.

The value of the radius of curvature is obtained from the expression:

$$R = \left[\left(\frac{b}{2} \right)^2 + a^2 \right] / 2a$$

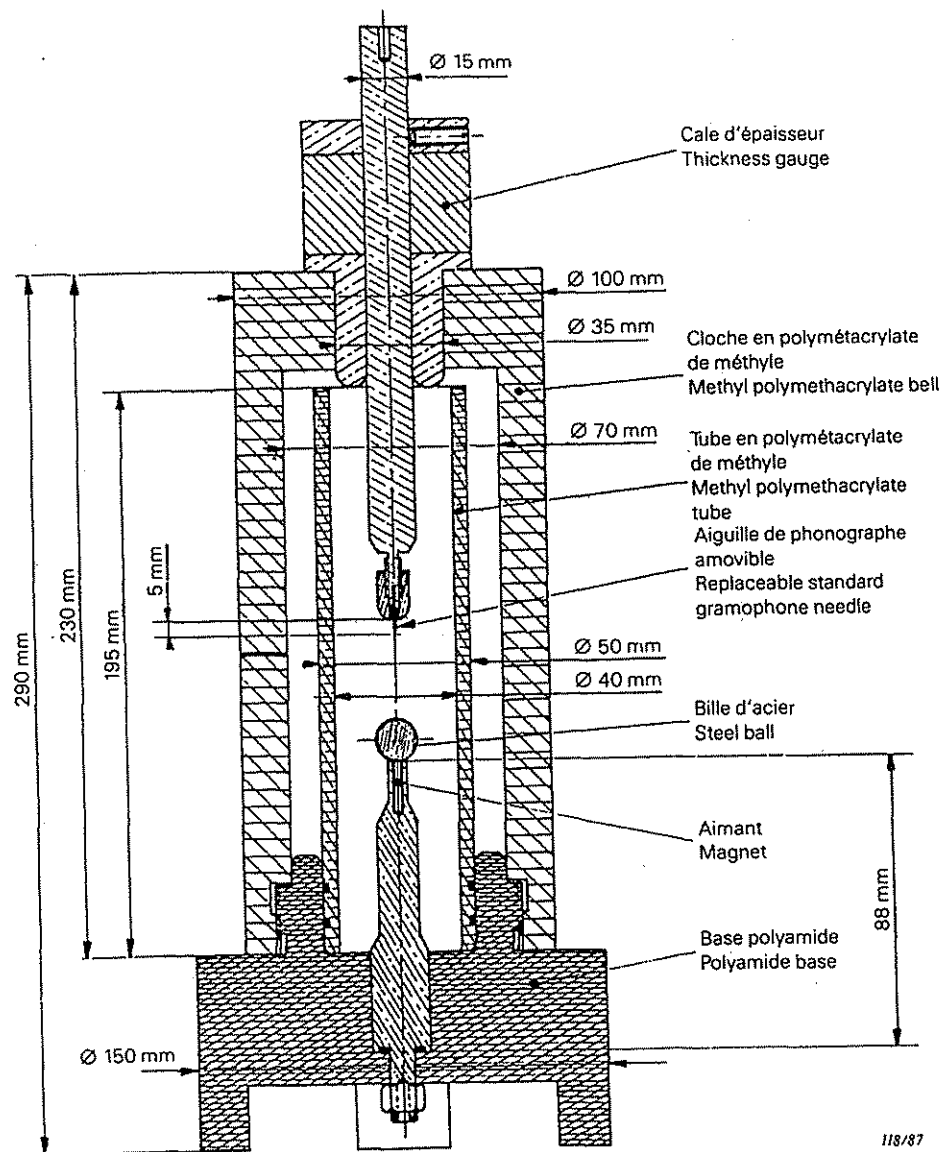
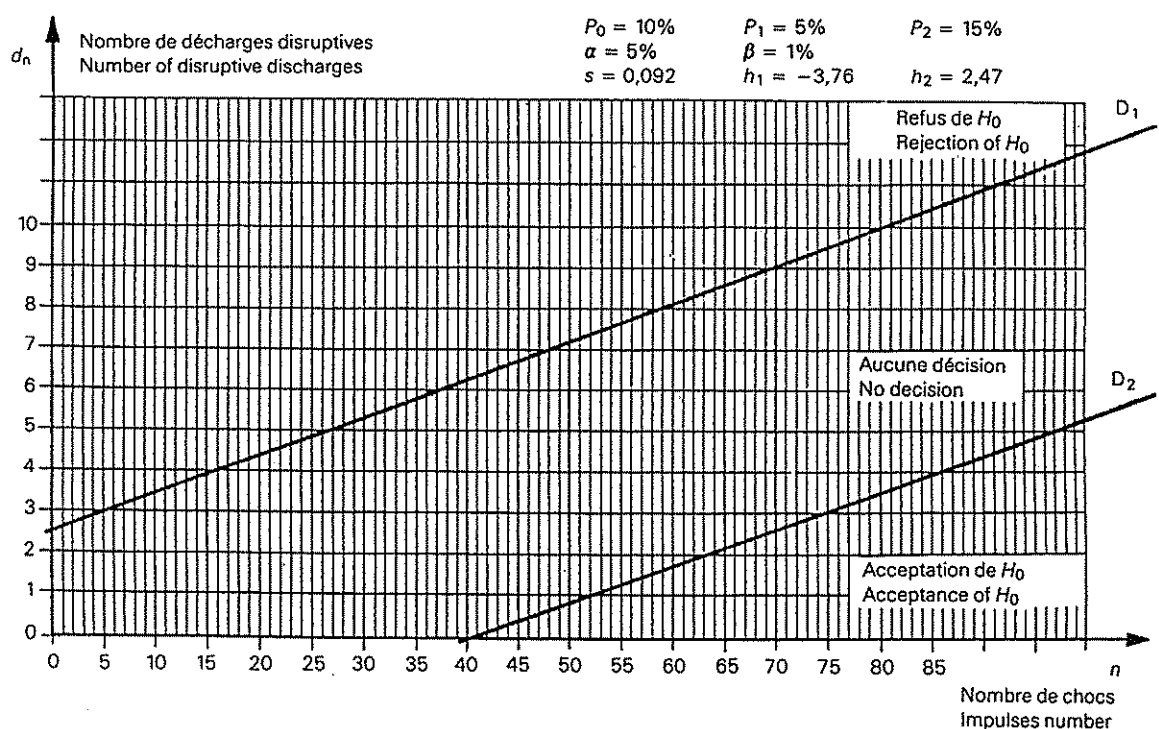
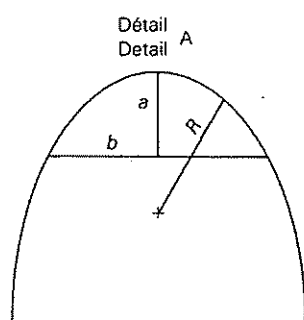


FIG. 1.— Système d'électrodes pointe-sphère pour essais de choc de liquides isolants.
Point-to-sphere electrode system for impulse strength of insulating liquids.

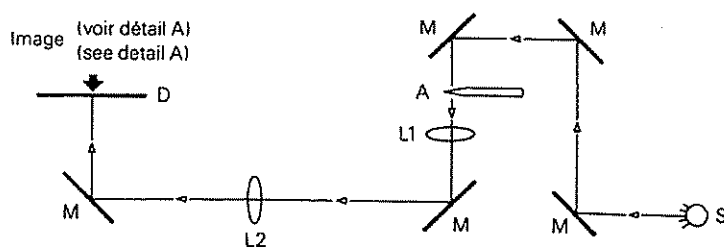


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FIG. 2.— Graphe de décision.
Ruling graph.



R = rayon de courbure
 radius of curvature
 a = flèche
 sagitta
 b = corde
 chord



S = source lumineuse
 source of light
 M = miroir
 mirror
 A = aiguille de phonographe
 gramophone needle

$L1$ = objectif grossissement 32x
 lens magnification 32x
 $L2$ = objectif grossissement 18x
 lens magnification 18x
 D = verre dépoli
 ground glass

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FIG. 3.— Mesurage du rayon de courbure des aiguilles.
Measurement of radius of curvature of needles.